

CO5-AT3

Questions & Viva Answers

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Q17. A web application is vulnerable to SQL injection. How would you evaluate the severity of this vulnerability and its possible consequences?

SQL Injection is an attack where malicious SQL commands are inserted into user input to manipulate a database. Test whether data can be read, modified, or deleted, whether login can be bypassed, and whether administrator access is possible. If sensitive data is exposed or the database can be controlled, the vulnerability is Critical because it affects confidentiality, integrity, and availability.

Q9. How would you evaluate the security risks of unnecessary services running on a Linux system?

Linux services are background programs that provide system or network functions. Check which services are running, whether they are required, whether they are exposed to the network, and whether they are up to date. Unnecessary services increase the attack surface and should be disabled to reduce security risk.

Q21. How would you evaluate the risk of hardcoded API keys or passwords discovered during mobile application analysis?

Hardcoded credentials are secrets stored directly inside the application's code instead of being securely protected. Verify whether the keys are active, what permissions they provide, and whether they can access backend services. The risk is High because attackers can extract the credentials and misuse the application's resources or data.

Q6. How would you distinguish between a vulnerable Windows service and a misconfigured Windows service during security testing?

A vulnerable service contains a software flaw that can be exploited, while a misconfigured service becomes insecure because of incorrect settings. Check the service version for known vulnerabilities and review permissions, authentication, and network exposure. Vulnerable services require patching, while misconfigured services require configuration changes.

Q25. You have identified vulnerabilities in Windows, Linux, messaging, web, and mobile applications. How would you prioritize them and prepare a professional security report?

Prioritize vulnerabilities based on severity, exploitability, business impact, and whether sensitive data is affected. Critical issues such as SQL Injection or exposed credentials should be fixed before medium or low-risk findings. A professional report should include an executive summary, affected systems, evidence, risk level, and recommended remediation for each vulnerability.